Royal Crest Academy
SCIENCE · INNOVATION · SKILL

FREE LEARNER GUIDE

Computer Science *study guide*

Computer Science teaches how computers work and how to instruct them, so you can design programs and systems that solve real problems.

Big ideas

Algorithms

An algorithm is a precise set of steps that solves a problem. Good algorithms are correct, clear and efficient.

Everything is binary

Computers store numbers, text, photos and sound as 0s and 1s. Eight bits make one byte.

Systems and networks

Hardware, software and networks work together. The internet moves data in small packets between connected devices.

Topic checklist

tick each topic when you can explain it

CORE TOPICS

- [ ] Computer systems and hardware
- [ ] Data representation: binary and hexadecimal
- [ ] Software and operating systems
- [ ] Algorithms and flowcharts
- [ ] Pseudocode and programming
- [ ] Variables, data types and selection
- [ ] Iteration (loops) and arrays
- [ ] Databases and SQL
- [ ] Networks and the internet
- [ ] Cyber security and online safety
- [ ] Logic gates
- [ ] Ethics and digital citizenship

GOING FURTHER

- [ ] Object-oriented programming
- [ ] Data structures: stacks, queues and lists
- [ ] Searching and sorting algorithms
- [ ] Web development: HTML, CSS and JavaScript
- [ ] Robotics and microcontrollers
- [ ] Introduction to artificial intelligence

Key terms

Algorithm	A step-by-step set of instructions for solving a problem.
Bit	A single binary digit: 0 or 1.
Byte	A group of 8 bits.
Variable	A named location in memory that stores a value which can change.
Iteration	Repeating a set of instructions, for example with a loop.
Debugging	Finding and fixing errors in a program.
Protocol	A set of rules for how data is sent across a network.
Encryption	Scrambling data so that only authorised people can read it.

In the computer lab

- Drawing flowcharts
- Writing pseudocode
- Programming in Python
- Tracing algorithms with trace tables
- Building simple web pages
- Coding robots in the Robotics & Coding club

Study smarter

1. Write a little code every day.
2. Trace an algorithm by hand before you run it.
3. Learn the standard algorithms: linear search, bubble sort.
4. Protect your accounts with strong passwords.

Check yourself

1. **Convert the binary number 1011 to denary.**

2. **How many bits are there in 3 bytes?**

3. **What does this output? total ← 0; FOR i ← 1 TO 4: total ← total + i; OUTPUT total**

Answers 1. 11 (8 + 0 + 2 + 1). · 2. 24 bits (3 × 8). · 3. 10 (1 + 2 + 3 + 4).

Where it leads: Software development, Cyber security, Data science, Network engineering, Robotics, Web and game design